Introduction to Machine Learning and Dataset Exploration

Learning Outcomes

At the end of this lab, students should be able to:

- Set up a Machine Learning development environment
- Understand the basic workflow of a Machine Learning project
- Load datasets using Pandas
- Explore datasets using Python
- Identify datasets, instances, attributes and labels
- Understand the Machine Learning concepts

Software Requirements

- Google Colab
- Jupyter Notebook
- VS Code with Python

Part A – Environment Setup

- Verify that Python is installed.
- Install the libraries (NumPy, Pandas, Scikit-Learn, etc.)
- Import the libraries

Part B – Working with a Dataset

1. Load the Iris dataset.
   (The Iris dataset is a built-in dataset available in the Scikit-learn library that contains measurements of 150 iris flowers from three species.)
2. Print number of rows, number of columns, feature names, target names.
3. Convert the dataset into a Pandas DataFrame.
4. Identify **Dataset, Instance, Attribute and Label** using the loaded dataset.
5. Explain why understanding the dataset is important before applying a machine learning algorithm.

Part C – Dataset Exploration

1. Display **shape, columns, data types, first and last few rows**, **information** and **describe** the dataset.
2. Based on the output obtained from **describe(),** answer the following questions.
   a. Which feature has the highest average value?
   b. Which feature has the largest variation (standard deviation)?
   c. Which feature has the smallest minimum value?

3. Display value counts for the target column.
4. Based on the output obtained above (4), answer the following questions.
   a. Is the Iris dataset balanced? Justify your answer.
   b. Why is checking the class distribution important before building a machine learning model?

Part D – Understanding Machine Learning Concepts

1. There are 3 common types of data.
   a. Using the examples below, identify the data type and justify your answer.
      - A customer database stored in MySQL
      - JSON data returned from a REST API
      - CCTV surveillance video
      - Emails containing headers and message bodies
      - An Excel spreadsheet containing student marks
   b. Which type of data is easiest to analyse using traditional machine learning techniques?
   c. Why is unstructured data more challenging than structured data?

2. For each scenario below, identify the appropriate learning paradigm.
   - Predicting whether a bank customer will default on a loan using historical labelled data
   - Grouping customers according to purchasing behavior without predefined labels
   - A robot learns to navigate a maze through rewards and penalties
   - Predicting house prices using historical housing data
   - Detecting unusual credit card transactions without labelled fraud cases

3. Three machine learning models were trained using the same dataset.

Model	Training Accuracy	Testing Accuracy
A	99%	61%
B	90%	88%
C	58%	55%

   a. Identify whether each model is,
      - Underfitting
      - Good Fit
      - Overfitting

      Provide a brief justification.
   b. What is the main disadvantage of an overfitted model?
   c. Suggest one technique that can help reduce overfitting.

4. Machine learning datasets are commonly divided into **Training, Validation, and Testing** sets before model development.
   a. Explain the primary purpose of the dataset splits.
   b. What are advantages of evaluating a machine learning model using a separate test dataset?
   c. Suppose a student trains and evaluates a model using the same dataset and obtains an accuracy of 99%. Would this result necessarily indicate that the model performs well on unseen data? Justify your answer.